

Lecture 3 - Block Diagrams

Instructor: Martin Nisser

Scribe: Catherine Ingram (Please send feedback to ingramca@uw.edu)

3.1 The 4 Control Objectives (Continued)

3.1.1 Sensitivity

We want to be robust to changes in the plant from G to $G + \Delta G$ because:

- The plant can change (e.g. friction).
- We might not have a perfect model G to begin with.

Specifically, we care how T (transfer function) will change due to a change in G (plant).

We can define,

$$\text{Sensitivity } S = \frac{\Delta T/T}{\Delta G/G} \quad \} \text{ The fractional change in } T \text{ given a fractional change in } G$$

In our original open-loop system, $T = GH$ and $\Delta T = (\Delta G)H$. Substituting this into S ,

$$S = \frac{\Delta T}{T} \cdot \frac{G}{\Delta G} = \frac{(\Delta G)H}{GH} \cdot \frac{G}{\Delta G} = 1$$

→ If G changes by 10%, then T changes by 10% (**Bad**).

But, in our closed-loop system, the derivation shows that,

$$S = \frac{1}{1 + GH} \quad \} \text{ This is why we called the } \frac{1}{1 + GH} \text{ term "S" in our output equation for } Y(s) \text{ earlier.}$$

→ If G changes by 10% ($\frac{\Delta G}{G} = \frac{1}{10}$) but $1 + GH = 100$, then T only changes by 0.1% ($\frac{\Delta T}{T} = S \cdot \frac{\Delta G}{G}$), this is **Good!**

∴ By choosing H to make $1 + GH$ large, our closed-loop system can be robust to plant changes.

3.1.2 Tracking

We want our output Y to track our reference R .

- A common metric is to evaluate the steady-state error.
→ We want $E_{ss} = 0$.
- More on this later.

3.2 Block Diagram Algebra

Recall:

- Lines act like signals $\rightarrow$
- Blocks act like amplifiers of signals (by value in block) $\square$
- Signals can be added/subtracted at junctions $\bigcirc \Sigma$

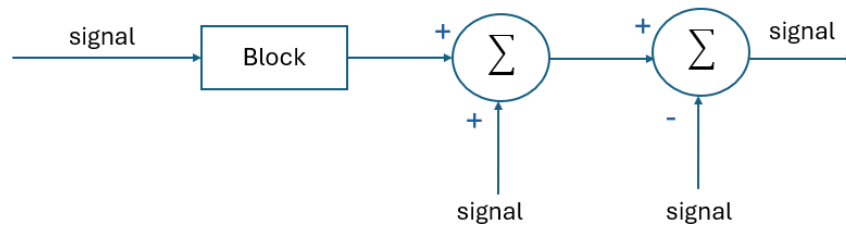


Figure 3.1: Example of block diagram components.

- Series Connections

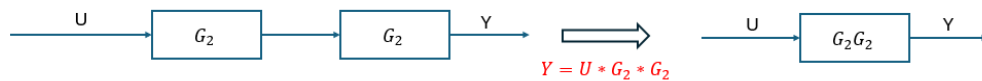


Figure 3.2: Series connections block diagram algebra.

- Parallel Connections (example shows addition, but can also do subtraction)

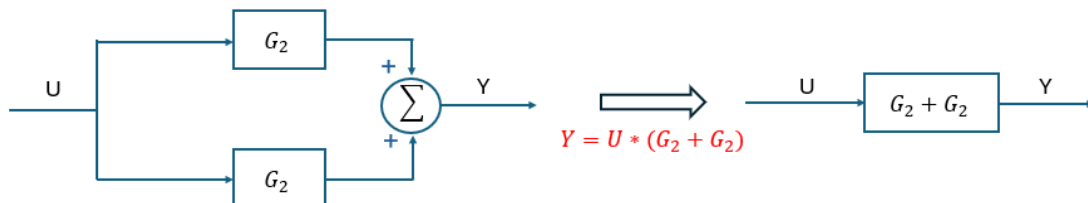


Figure 3.3: Parallel connections block diagram algebra.

- Feedback (example shows subtraction, but can also do addition)

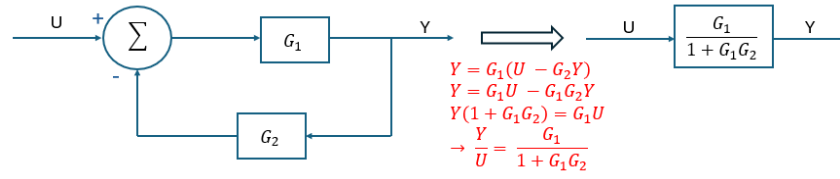


Figure 3.4: Feedback system block diagram.

Example

System with multiple external inputs: U (control input) and W (disturbance)

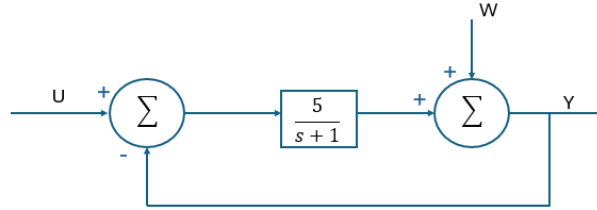


Figure 3.5: Feedback system with control and disturbance inputs.

The equation for the output $Y(s)$ is the superposition of its response to each external input:

$$Y = G_{UY}U + G_{WY}W$$

What are the transfer functions G_{UY} ($U \rightarrow Y$) and G_{WY} ($W \rightarrow Y$)?

1. G_{UY} : Let $W = 0$,

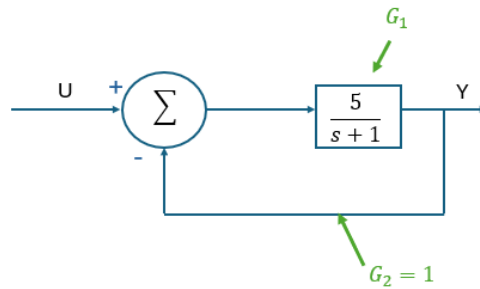
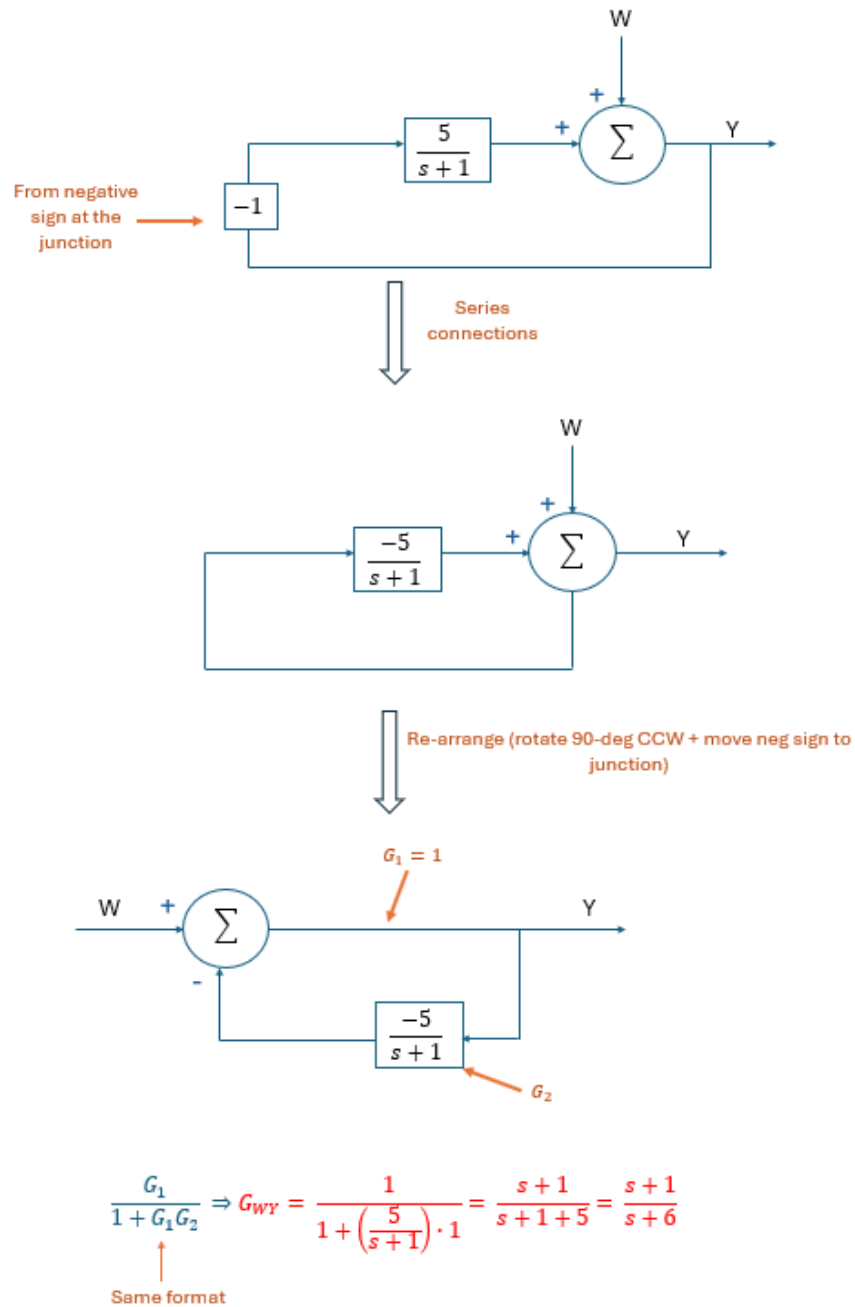


Figure 3.6: Setting $W = 0$. Mirroring the feedback example in Fig. 3.4, treat this as $G_2 = 1$ or "unity feedback".

$$\text{Recall: } \underbrace{G_{UY} = \frac{G_1}{1 + G_1 G_2}}_{\text{From feedback e.g. above}} \implies G_{UY} = \frac{\frac{5}{s+1}}{1 + \frac{5}{s+1}(1)} = \frac{5}{s+1+5} = \frac{5}{s+6}$$

2. G_{WY} : Let $U = 0$,

Figure 3.7: Setting $U=0$, we then simplify the block diagram in sequential steps.

Example (same problem viewed from the basics)

If in doubt, go back to the basics,

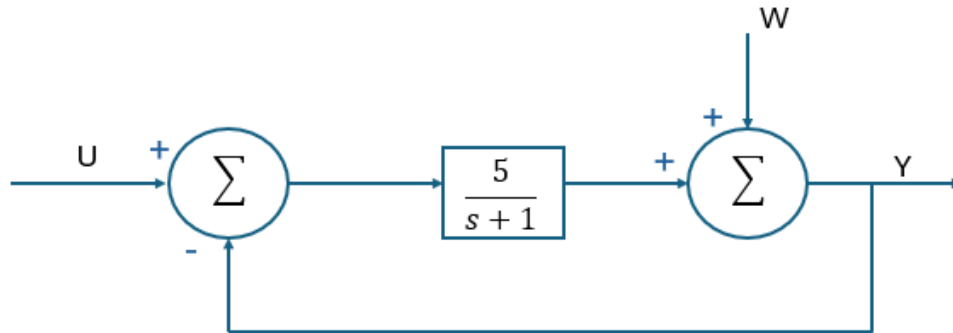


Figure 3.8: Same problem as before.

$$Y = \frac{5}{s+1}(U - Y) + W$$

$$Y + \frac{5}{s+1}Y = \frac{5}{s+1}U + W$$

$$Y = \frac{\frac{5}{s+1}}{1 + \frac{5}{s+1}}U + \frac{1}{1 + \frac{5}{s+1}}W$$

$$Y = \frac{5}{s+1+5}U + \frac{s+1}{s+1+5}W$$

$$\therefore Y = \frac{5}{s+6}U + \frac{s+1}{s+6}W$$